ENGINEERING SOFTWARE DEVELOPMENT | REVISED PROJECT BRIEF

Aircraft Flight Performance Simulator

CONTRACT: AERO-FPS-2026-001 **REVISION:** 02 / ML Extension

Problem statement. Develop a Python-based engineering application that estimates aircraft performance from the start of the takeoff roll through landing. The application shall combine simplified flight-physics models with one small machine-learning exercise, supporting repeatable simulations, configuration comparisons, and clear presentation of results.

01 | Simulation Scope

- **Aircraft definition.** Allow users to specify representative aircraft mass, geometry, aerodynamic characteristics, propulsion capability, and fuel load.
- **Operating environment.** Account for airport elevation, atmospheric conditions, wind, and runway characteristics relevant to the mission.
- **Flight phases.** Simulate takeoff ground roll, liftoff and climb, cruise, descent, approach, and landing rollout; track flight state and fuel consumption over time.
- **Engineering outputs.** Report takeoff and landing distances, climb and cruise performance, estimated range and endurance, and meaningful plots of the flight profile.
- **Verification.** Perform reasonableness checks and compare selected results with published aircraft-performance references or clearly documented benchmarks.

02 | Single Machine-Learning Work Package

Task: Develop one introductory supervised-regression model that predicts **takeoff ground-roll distance** from aircraft and operating-condition inputs. This is the project's **only required ML model**.

- Create a labeled dataset by running the physics-based ground-roll simulation across varied aircraft weight, thrust capability, wind, atmospheric conditions, and runway slope.
- Select suitable input variables, separate training and held-out test cases, and train one basic regression model.
- Evaluate prediction error on unseen scenarios and show predicted versus simulated takeoff distances.
- Display the ML estimate beside the physics-based result; keep the primary flight simulation functional without the ML model.

03 | Modeling Boundaries

Use an aircraft-level, preliminary-performance fidelity appropriate for a standalone Python project. Full computational fluid dynamics, detailed engine-internal aerodynamics, and flight-control-system design are outside scope. The ML model is a **surrogate for the simulation data**, not an independently validated predictor of real-world aircraft performance.

04 | Required Deliverables

A working Python project with modular source code; a documented aircraft/configuration input method; reproducible example missions; flight-performance tables and plots; one trained ML model with its training and test results; an ML-versus-simulation comparison; and a short report explaining assumptions, accuracy, and limitations.